JATROPHA AGROFORESTRY SENEGAL

MONITORING REPORT

03/06/2009 – 02/06/2013

Document Prepared By CarbonSinkGroup s.r.l.

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Table of Contents

1 PROJECT DETAILS

1.1 Summary Description of Project

The Jatropha Agroforestry Senegal project develops Jatropha curcas plantations in the Fatick, Kaolack and Kaffrine regions, in central-western agricultural region of the Senegal. The Jatropha curcas plantations are foreseen cover an area of 1411.34 ha of degraded soils mainly in the surroundings of the cities of Ourour and Kaffrine. The plantations will be located near villages around these cities to limit transport and facilitate the access of inhabitants as the project involves local hand-workers from villages. Local workers will be trained for both sustainable agricultural and forestry practices and hired for all plantation operational activities.

The planned agro-forestry system will assure a proper management of trees in cultivated lands allowing the carbon storage by plants biomass and soils, consequently providing a contribution to the greenhouse effect mitigation and the reversal of soil degradation trend.

African National Oil Corporation s.a.r.l. (ANOC) has concessions for the project lands and will be the owner of Jatropha plants and products. Moreover, ANOC will manage the agricultural operations (e.g., sowing, fertilizing). In addition, contemporary crop activities will be boosted as local farmers will be working only part-time in the Jatropha fields. The unused fallow agricultural land, where Jatropha will be grown, were legally assigned to ANOC for 25% of the available agricultural land, hence ensuring to continue to produce both food and traditional agricultural products (e.g., peanuts) as usual in the baseline scenario.

The aims of this project are:

- The sequestration of carbon dioxide (CO_2) through the nurturing of Jatropha curcas.
- The regeneration of degraded soils, protecting them against erosion processes and saving existing carbon stocks.
- The empowerment of local communities to develop sustainable agro-forestry practices.
- The provision of rotated livelihood potentialities for local communities with the possible access to alternative income for local stakeholders and the promotion of sustainable agricultural practices (e.g., Jatropha farming will harvest fruit whose seeds can be sold by villagers so as to be mechanically pressed to extract oil and generate the biodiesel fuel).
- The local production of biodiesel from Jatropha seeds will replace the scarcely-available diesel fuel, which is currently used to generate electricity; consequently, the biodiesel will provide an alternative source of energy for the sustainable development of this region.

1.2 Sectoral Scope and Project Type

The project is a Voluntary Carbon Standard (VCS) project within the sectoral scope of Agriculture, Forestry and Other Land Use (AFOLU) and category Afforestation, Reforestation and Revegetation (ARR). The project is a grouped project.

1.3 Project Proponent

African National Oil Corporation s.a.r.l. (ANOC) is the Coordinating/Managing Entity (C/ME) of the project. ANOC is a Senegalese private company owned by the Italian Bioenergy Production s.r.l. In Senegal its activities started in 2008 with the development of first nurseries in several villages and made first pilot Jatropha plantations near Ourour (Fatic region) and Kaffrine (Kaffrine region). ANOC developed the plantations respecting the environment by using sustainable agricultural practices (e.g., green manure, no mechanization), consistently with national and local laws, and cooperating with the Biofuels Division of the Senegal Ministry of Energy and Biofuels.

Contact information:

Organization:	African National Oil Corporation s.a.r.l.
Street/P.O.Box:	Ngaparou
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City:	Mbour
Country:	Senegal
Telephone:	+221 774531466
Represented by:	Managing Director
Last Name:	Milani
First Name:	Alessandro
Mobile:	+39 388 9373256
Personal E-Mail:	studio_milani@libero.it

Roles/responsibilities:

ANOC is the project proponent and managing entity. ANOC will manage the project activity (like agricultural operations e.g., sowing, fertilizing) and the monitor activities.

1.4 Other Entities Involved in the Project

Agroils Technologies s.r.l.

Agroils Technologies s.r.l. has been formed to provide technical assistance to Jatropha producers. Today Agroils is the European leader in Jatropha consultancy and it is a promoting partner of the Jatrophabook ONLUS, the most important stakeholders' platform in the sector. Agroils has developed a strong network with top research institutes (IFEU, Hohenheim, Wageningen) and is an active player in the international debate on sustainability (GBEP, RSB). In addition, recently Agroils has become operational as a technical consultant in the carbon sector both for voluntary and mandatory initiatives, with special focus on biofuels and AFOLU projects.

Contact information:

Organization:	Agroils Technologies s.r.l.
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Last Name:	Venturini Del Greco
First Name:	Giovanni
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Personal E-Mail:	giovannivdg@agroils.com

Roles/responsibilities:

The role of Agroils Technologies s.r.l. is to give technical consultation for the project.

CarbonSinkGoup s.r.l

Carbon Sink Group s.r.l. is a specialized consultancy company dealing with the environmental preservation and management. Thanks to the recognition as Academic Spin-Off by the University of Florence and the professional background of its team, Carbon Sink Group is focused on projects aimed to the reduction and offsetting of the greenhouse gas (GHGs) emission in the atmosphere.

Contact information:

Organization:	CarbonSinkGroup s.r.l.
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Last Name:	Maggiani
First Name:	Andrea
Mobile:	+39 3407109826
Personal E-Mail:	andrea@carbonsink.it

Roles/responsibilities:

The role of CarbonSinkGroup s.r.l is to give technical and scientific consultation in the phase of preparation of the project design document, validation, monitoring and verification as well as on commercializing the produced credits.

1.5 Project Start Date

The starting date of the project is 03 June 2009.

1.6 Project Crediting Period

The project crediting period will be 24 years and 06 months. The crediting period starting date is 03 June 2009 and the ending date is 31 December 2033.

1.7 Project Location

The grouped project boundary is the boundaries of the regions of Fatick, Kaolack and Kaffrine (Figure 1). All the project activity instances will be within these regions.

All the plantation plots of the first project instance are located in the vicinity of the major cities of Ourour and Kaffrine, within regions of Fatick, Kaolack and Kaffrine (Figure 2 and Table 1). More detailed boundary of the project area included in this monitoring report (the first project instance) is presented in separate file in Google Earth KML format[1].

Figure 1. Regions of Senegal (source: Wikipedia[2])

[1] KML_Plantations_2009_2010_2011
[2] Wikipedia: http://en.wikipedia.org/wiki/File:Regions_of_Senegal.svg. Site visited 17/04/2013.

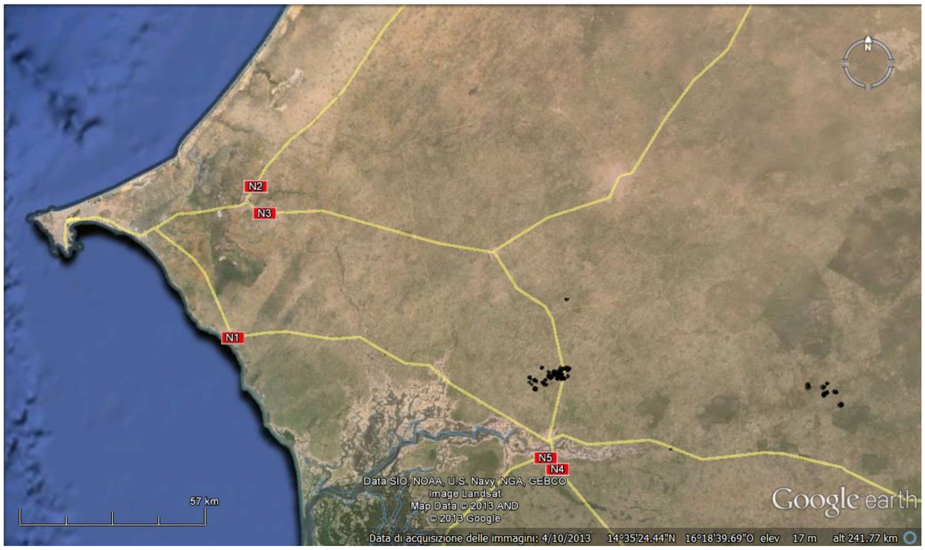

Figure 2. Location of the first project instance (source: Google Earth)

Table 1. Project area plots of the first project instance

Name	Location (Region)	Area (ha) and planting year		
		2009	2010	2011
OUROUR FORAGE 19	Fatick	9.19		
OUROUR CADA 20	Fatick	3.12		
OUROUR CADA 21	Fatick	9.00		
OUROUR CAMARA 24	Fatick	1.77		
OUROUR CAMARA 25	Fatick	17.40		
KEUR MIGNAN 12	Kaolack	1.77		
KEUR MIGNAN 13	Kaolack	3.00		
KEUR MIGNAN 14	Kaolack	2.77		
NAWEL 1	Kaolack	1.11		
OUROUR 1	Fatick		5.01	
OUROUR 2	Fatick		3.59	
OUROUR 3	Fatick		6.25	
OUROUR 4	Fatick		4.98	
OUROUR 5	Fatick		9.87	
OUROUR 6	Fatick		33.5	
OUROUR 7	Fatick		14.40	
OUROUR 8	Fatick		2.75	
COLOMBANE 2	Fatick		5.79	
SOUMBEL 1	Fatick		12.90	
SOUMBEL 2	Fatick		6.63	
SOUMBEL NUOVO 3	Fatick		3.26	
COLOMBANE 1	Fatick		3.00	
COLOMBANE NUOVO 3	Fatick		3.71	
KEUR DIEGANE 1	Fatick		1.97	
KEUR DIEGANE 2	Fatick		5.78	
KEUR DIEGANE 3	Fatick		4.64	
KEUR MIGNAN 2	Kaolack		1.89	
OUROUR 22	Fatick		2.10	

NDIOTE MOR 1	Kaffirine		32.00	
NDIOTE MOR 2	Kaffirine		12.09	
NDIOLEFENE 1	Kaffirine		19.50	
DELBY 1	Kaffirine		24.90	
DIANKE KAO 1	Kaffirine		17.80	
OUROUR CRISE 10	Fatick			26.01
OUROUR VIRAGE BAC 15	Fatick			1.56
OUROUR ROUTE DE GAGNIK 16	Fatick			4.00
ROUTE DE SOUMBEL 1	Fatick			6.68
ROUTE DE SOUMBEL 2	Fatick			0.44
ROUTE DE SOUMBEL 3	Fatick			16.70
ROUTE DE SOUMBEL 4	Fatick			2.52
ROUTE DE SOUMBEL 5	Fatick			2.71
OUROUR 13	Fatick			29.00
OUROUR 14	Fatick			3.60
MAKA SOUMBEL 1	Fatick			3.28
MAKA SOUMBEL 2	Fatick			6.85
SOUMBEL GALLO 1	Fatick			2.95
SOUMBEL GALLO 2	Fatick			3.90
SOUMBEL GALLO 3	Fatick			2.20
SOUMBEL GALLO NEW 4	Fatick			2.77
GOWETHE SERERE 1	Fatick			2.00
GOWETHE SERERE 3	Fatick			5.61
COLOBANE 4	Fatick			12.20
KEUR MIGNANE DIOP 5	Kaolack			4.42
KEUR DIEGANE NEW 7	Fatick			2.42
KEUR MIGNANE 4	Kaolack			0.95
KEUR MIGNANE 6	Kaolack			1.89
GOWETHE WADENE 1	Fatick			4.86
KEUR MIGNANE 7	Kaolack			2.96
KEUR MIGNANE 8	Kaolack			0.35
KEUR MIGNANE 9	Kaolack			0.44
KEUR MIGNANE 10	Kaolack			0.78

TEOWROUCOSOM 1	Fatick			8.84
DIANKE SOUF 1	Kaffrine			40.02
DELBY 2	Kaffrine			25.30
NDIOTE MOR ENTREE 1	Kaffrine			24.40
NDIOTE MOR ENTREE 3	Kaffrine			11.00
NDIOLOFEME 2	Kaffrine			12.50
NDIAAW NEW 1	Kaffrine			7.78
Total area (ha)		**49.13**	**238.31**	**283.89**

1.8 Title and Reference of Methodology

The project will use an UNFCCC approved CDM A/R baseline and monitoring methodology, called AR-ACM0003 "A/R Large-scale Methodology: Afforestation and reforestation of lands except wetlands" (Version 01.0.0).

The methodology is available at UNFCCC website:
http://cdm.unfccc.int/methodologies/ARmethodologies/approved

2 IMPLEMENTATION STATUS

2.1 Implementation Status of the Project Activity

This monitoring period, referencing the project years 1, 2, 3 and 4 (equivalent 03 June 2009 to 02 June 2013). This period includes project activities implemented since the project start date, but prior to the validation of the project. During this monitoring period ANOC has undertaken forestation activities (i.e. plantation establishment and management activities) as described in the project design document (PD). The project has undertaken all required monitoring activities as outlined in Section 3.3 of this document.

Permanence is addressed through the risk buffer. In accordance with the VCS Standard (version 3.3) a non-permanence risk analysis has been prepared in accordance with AFOLU Non-Permanence Risk Tool[3]. As a result of the risk analysis, the project will hold a 37 % VCS buffer in reserve for use in case of any loss of permanence. The risk analysis is available in a separate file called "Non_Permanence_Risk_Report[4]".

[3] AFOLU Non-Permanence Risk Tool v.3.2
http://v-c-s.org/sites/v-c-s.org/files/AFOLU%20Non-Permanence%20Risk%20Tool%2C%20v3.2.pdf
[4] Non-Permanence Risk Report

2.2 Project Description Deviations

The project and project monitoring plan meets all of the requirements of the applied methodology and does not deviate from the baseline scenario, additionality determination, or inclusion of project GHG sources, sinks and reservoirs.

2.3 Grouped Project

During the first monitoring period there was not any new instances added to the project activity.

3 DATA AND PARAMETERS

3.1 Data and Parameters Available at Validation

Data Unit / Parameter:	CF_{TREE_BSL}
Data unit:	t C (t.d.m.)$^{-1}$
Description:	Carbon fraction of tree biomass in the baseline
Source of data:	Default value[5]
Value applied:	0.47
Purpose of the data:	Calculation of baseline emissions or baseline net GHG removals by sinks
Any comment:	N/A

Data Unit / Parameter:	B_{FOREST}
Data unit:	t d.m. ha^{-1}
Description:	Default above-ground biomass content in forest in the region/country where the A/R project is located
Source of data:	Table 3A.1.4 of IPCC GPG-LULUCF 2003/Annex 3A.1 (value for Senegal)[6]
Value applied:	30
Purpose of the data:	Calculation of baseline emissions or baseline net GHG removals by sinks
Any comment:	N/A

[5] AR-TOOL14. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-14-v3.0.0.pdf
[6] IPCC GPG-LULUCF 2003. http://www.ipcc-nggip.iges.or.jp/public/gpglulucf/gpglulucf.html (Site visited 28.11.2013)

Data Unit / Parameter:	R_{TREE_BSL}
Data unit:	Dimensionless
Description:	Root-shoot ratio for the trees in baseline
Source of data:	Default value[7]
Value applied:	0.25
Purpose of the data:	Calculation of baseline emissions or baseline net GHG removals by sinks
Any comment:	N/A

Data Unit / Parameter:	ΔB_{FOREST}
Data unit:	t d.m. ha^{-1} yr^{-1}
Description:	Default average annual increment in above-ground biomass in forest in the region/country where the A/R project is located
Source of data:	Table 3A.1.5 of IPCC GPG-LULUCF 2003/Annex 3A.1. (value for Africa, dry climate)[8]
Value applied:	For the years 1-20: 1.2 For the years 21-25: 0
Purpose of the data:	Calculation of baseline emissions or baseline net GHG removals by sinks
Any comment:	(a) Tree biomass may reach a steady state when biomass growth becomes zero or insignificant – either because of biological maturity of trees or because the rate of anthropogenic biomass extraction from the area is equal to the rate of biomass growth. Therefore, this parameter should be taken to be zero after the year in which tree biomass in baseline reaches a steady state. The year in which tree biomass in baseline reaches steady-state is taken to be the 20th year from the start of the project activity, unless transparent and verifiable information can be provided to justify a different year; (b) When land is subjected to periodic slash-and-burn practices in the baseline, the average tree biomass is constant, and hence value of this parameter is set equal to zero.

[7] AR-TOOL14. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-14-v3.0.0.pdf
[8] IPCC GPG-LULUCF 2003. http://www.ipcc-nggip.iges.or.jp/public/gpglulucf/gpglulucf.html (Site visited 28.11.2013)

Data Unit / Parameter:	**CF_S**
Data unit:	t C (t.d.m.)$^{-1}$
Description:	Carbon fraction of shrub biomass
Source of data:	IPCC default value
Value applied:	0.47
Purpose of the data:	Calculation of baseline emissions or baseline net GHG removals by sinks
Any comment:	N/A

Data Unit / Parameter:	**R_s**
Data unit:	Dimensionless
Description:	Root-shoot ratio for shrubs
Source of data:	Table 4.4 of 2006 IPCC Guidelines for National Greenhouse Gas Inventories[9]
Value applied:	0.40
Purpose of the data:	Calculation of baseline emissions or baseline net GHG removals by sinks
Any comment:	The value of R_S shall be 0.40 [Table 4.4 of 2006 IPCC Guidelines for National GHG Inventories] unless transparent and verifiable information can be provided to justify different values[10]

Data Unit / Parameter:	**BDR_{SF}**
Data unit:	Dimensionless
Description:	Ratio of biomass per hectare in land having a shrub crown cover of 1.0 (i.e. 100%) and the default above-ground biomass content in forest in the region/country where the A/R project is located
Source of data:	Default value[11]
Value applied:	0.10
Purpose of the data:	Calculation of baseline emissions or baseline net GHG removals by sinks
Any comment:	N/A

[9] IPCC Guidelines for National GHG Inventories. http://www.ipcc-nggip.iges.or.jp/public/2006gl/ (Site visited 28.11.2013)
[10] AR-TOOL14. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-14-v3.0.0.pdf
[11] AR-TOOL14. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-14-v3.0.0.pdf

Data Unit / Parameter:	R_j
Data unit:	Dimensionless
Description:	Root-shoot ratio for tree species j
Source of data:	Calculated
Value applied:	See excel-file “Calculations ex post first instance”[12]
Purpose of the data:	Calculation of project emissions or actual net GHG removals by sinks
Any comment:	The value of Rj is calculated as R = exp[-1.085+0.9256*ln(A)]/A, where A is above-ground biomass content (t d.m. ha-1) [Table 4.A.4 of IPCC GPG LULUCF 2003] unless transparent and verifiable information can be provided to justify different values.

Data Unit / Parameter:	CF_{TREE}
Data unit:	Carbon fraction of tree biomass
Description:	Carbon fraction of dry matter for species or group of species j
Source of data:	Default value[13]
Value applied:	0.47
Purpose of the data:	Calculation of project emissions or actual net GHG removals by sinks
Any comment:	N/A

[12] Calculations ex post first instance
[13] AR-TOOL14. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-14-v3.0.0.pdf

Data Unit / Parameter:	**$f_j(x1_{p,i,t}, x2_{p,i,t}, x3_{p,i,t}, \ldots)$**
Data unit:	t d.m.
Description:	Function relating measured tree dimensions (x1, x2, x3, …) to above-ground tree biomass
Source of data:	Equation of Ghezehei et al. (2009)[14] modified with a multiplier
Value applied:	Total above ground biomass (g d.m.) = 1.213*0.000907*stem base diameter (mm)$^{3.354}$
Purpose of the data:	Calculation of project emissions or actual net GHG removals by sinks
Any comment:	The demonstration of the appropriatenes of the selected equation is demonstrated in a separate document “Demonstrating appropriateness of allometric equation for the project Jatropha Agroforestry Senegal”[15]

Data Unit / Parameter:	**t_{VAL}**
Data unit:	dimensionless
Description:	Two-sided Student’s t-value, at infinite degrees of freedom in the first iteration and at degrees of freedom equal to (n-1) in subsequent iterations, for the required confidence level; dimensionless
Source of data:	Student’s t-distribution table available in the tool “Calculation of the number of sample plots for measurements within A/R CDM project activities”[16]
Value applied:	1.96
Purpose of the data:	Calculation of project emissions or actual net GHG removals by sinks (calculating the number of sample plots)
Any comment:	N/A

[14] Ghezehei B, Annandale JG and Everson CD (2009), Shoot allometry of Jatropha curcas, Southern Forests, 2009.
[15] Demonstrating appropriateness of allometric equation for the project “Jatropha Agroforestry Senegal”.CarbonSinkGroup, 2013.
[16] EB 58, Annex 15. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-03-v2.1.0.pdf

Data Unit / Parameter:	**E**
Data unit:	t d.m. ha^{-1}
Description:	Acceptable margin of error (i.e. one-half the confidence interval) in estimation of biomass stock within the project boundary; in units used for s_i.
Source of data:	Calculated
Value applied:	3.06
Purpose of the data:	Calculation of project emissions or actual net GHG removals by sinks (calculating the number of sample plots)
Any comment:	The selected value is calculated as 15% of the mean biomass stock within the project boundary. See Excel-file ”Sample plot size”[17]

Data Unit / Parameter:	**$SOC_{REF,i}$**
Data unit:	t C ha^{-1}
Description:	Reference SOC stock corresponding to the reference condition in native lands (i.e. non-degraded, unimproved lands under native vegetation – normally forest) by climate region and soil type applicable to stratum i of the areas of land
Source of data:	Default value[18]
Value applied:	31
Purpose of the data:	Calculation of project emissions or actual net GHG removals by sinks
Any comment:	SOC it is not accounted for the first project instance

[17] See Excel-file “Sample_plot_size”
[18] EB 60, Annex 12. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-16-v1.1.0.pdf

Data Unit / Parameter:	$f_{LU,i}$
Data unit:	dimensionless
Description:	Relative stock change factor for baseline land-use in stratum i of the areas of land
Source of data:	Default value[19]
Value applied:	1.00
Purpose of the data:	Calculation of project emissions or actual net GHG removals by sinks
Any comment:	SOC it is not accounted for the first project instance

Data Unit / Parameter:	$f_{MG,i}$
Data unit:	dimensionless
Description:	Relative stock change factor for baseline management regime in stratum i of the areas of land
Source of data:	Default value[20]
Value applied:	0.97
Purpose of the data:	Calculation of project emissions or actual net GHG removals by sinks
Any comment:	SOC it is not accounted for the first project instance

Data Unit / Parameter:	$f_{IN,i}$
Data unit:	dimensionless
Description:	Relative stock change factor for baseline input regime (e.g. crop residue returns, manure) in stratum i of the areas of land
Source of data:	Default value[21]
Value applied:	1.00
Purpose of the data:	Calculation of project emissions or actual net GHG removals by sinks
Any comment:	SOC it is not accounted for the first project instance

[19] EB 60, Annex 12. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-16-v1.1.0.pdf
[20] EB 60, Annex 12. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-16-v1.1.0.pdf
[21] EB 60, Annex 12. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-16-v1.1.0.pdf

3.2 Data and Parameters Monitored

<table>
<tr><td>Data Unit / Parameter:</td><td>$A_{BSL,i}$</td></tr>
<tr><td>Data unit:</td><td>Ha</td></tr>
<tr><td>Description:</td><td>Area of stratum i in the baseline, delineated on the basis of tree crown cover at the start of the A/R project activity; ha</td></tr>
<tr><td>Source of data:</td><td>Field measurement</td></tr>
<tr><td>Description of measurement methods and procedures to be applied:</td><td>Standard operating procedures (SOPs) prescribed under national forest inventory are applied. In the absence of these, SOPs from published handbooks, or from the IPCC GPG LULUCF 2003, are applied</td></tr>
<tr><td>Frequency of monitoring/recording:</td><td>Every five years since the year of the initial verification</td></tr>
<tr><td>Value monitored:</td><td>For the first project instance, the baseline situation:
<table>
<tr><td></td><td>Area (ha)</td></tr>
<tr><td>Tree crown cover stratum</td><td>First instance</td></tr>
<tr><td>Stratum 1</td><td>571.33[22]</td></tr>
</table></td></tr>
<tr><td>Monitoring equipment:</td><td>N/A</td></tr>
<tr><td>QA/QC procedures to be applied:</td><td>Quality control/quality assurance (QA/QC) procedures prescribed under national forest inventory are applied. In the absence of these, QA/QC procedures from published handbooks, or from the IPCC GPG LULUCF 2003, are applied</td></tr>
<tr><td>Calculation method:</td><td>N/A</td></tr>
<tr><td>Any comment:</td><td>N/A</td></tr>
</table>

[22] The borders are equal to the borders of the first project instance presented in KML_Plantations_2009-2010-2011

Data Unit / Parameter:	$A_{SHRUB,i,t}$
Data unit:	Ha
Description:	Area of shrub biomass stratum i at a given point of time in year t
Source of data:	Field measurement
Description of measurement methods and procedures to be applied:	Standard operating procedures (SOPs) prescribed under national forest inventory are applied. In the absence of these, SOPs from published handbooks, or from the IPCC GPG LULUCF 2003, are applied
Frequency of monitoring/recording:	Every five years since the year of the initial verification
Value monitored:	For the first project instance, the baseline situation[23]: <table><tr><td></td><td>Area (ha)</td></tr><tr><td>Shrub stratum</td><td>First instance</td></tr><tr><td>Stratum 1</td><td>344.04</td></tr><tr><td>Stratum 2</td><td>227.29</td></tr></table> For the first project instance, the project situation[24]: <table><tr><td></td><td>Area (ha)</td></tr><tr><td>Shrub stratum</td><td>First instance</td></tr><tr><td>Stratum 1</td><td>571.33</td></tr></table>
Monitoring equipment:	N/A
QA/QC procedures to be applied:	Quality control/quality assurance (QA/QC) procedures prescribed under national forest inventory are applied. In the absence of these, QA/QC procedures from published handbooks, or from the IPCC GPG LULUCF 2003, are applied
Calculation method:	N/A
Any comment:	N/A

23 KML_Baseline Shrub Stratum1 and KML_Baseline_Shrub_Stratum2
24 The borders are equal to the borders of the first project instance presented in KML_Plantations_2009-2010-2011

Data Unit / Parameter:	**$CC_{TREE\_BSL,i}$**
Data unit:	Dimensionless
Description:	Crown cover of trees in the baseline, in baseline stratum i, expressed as a fraction (e.g. 10% crown cover implies $CC_{TREE\_BSL,i}$ =0.10)
Source of data:	Field observation
Description of measurement methods and procedures to be applied:	Considering that the biomass in trees in the baseline is smaller compared to the biomass in trees in the project, a simplified method of measurement may be used for estimating tree crown cover. Ocular estimation of tree crown cover may be carried out or any other method such as the line transect method or the relascope method may be applied
Frequency of monitoring/recording:	Every five years since the year of the initial verification and as well at the time of planting of new project instances
Value monitored:	For the first project instance: 0.01
Monitoring equipment:	N/A
QA/QC procedures to be applied:	Quality control/quality assurance (QA/QC) procedures prescribed under national forest inventory are applied. In the absence of these, QA/QC procedures from published handbooks, or from the IPCC GPG LULUCF 2003, are applied
Calculation method:	N/A
Any comment:	Only one baseline tree crown cover stratum has been identified.

Data Unit / Parameter:	$CC_{SHRUB,i,t}$
Data unit:	Dimensionless
Description:	Crown cover of shrubs in shrub biomass stratum i at a given point of time in year t
Source of data:	Field observation
Description of measurement methods and procedures to be applied:	Considering that the biomass in shrubs is smaller than the biomass in trees, a simplified method of measurement may be used for estimating shrub crown cover. Ocular estimation of crown cover

	may be carried out or any other method such as the line transect method or the relascope method may be applied
Frequency of monitoring/recording:	Every five years since the year of the initial verification and as well at the time of planting of new project instances
Value monitored:	For the first project instance, the baseline situation: <table><tr><td>Shrub stratum</td><td>$CC_{SHRUB,t}$</td></tr><tr><td>Stratum 1</td><td>0.05*</td></tr><tr><td>Stratum 2</td><td>0.10*</td></tr></table>*one half of observed maximum shrub crown cover For the first project instance, the project situation: <table><tr><td>Shrub stratum</td><td>$CC_{SHRUB,t}$</td></tr><tr><td>Stratum 1</td><td>0</td></tr></table>
Monitoring equipment:	N/A
QA/QC procedures to be applied:	Quality control/quality assurance (QA/QC) procedures prescribed under national forest inventory are applied. In the absence of these, QA/QC procedures from published handbooks, or from the IPCC GPG LULUCF 2003, are applied
Calculation method:	N/A
Any comment:	(a) When land is subjected to periodic slash-and-burn practices in the baseline, an average shrub crown cover equal to default value of 0.5 is used in Equation (35) unless transparent and verifiable information can be provided to justify a different value; The baseline shrub cover was confirmed from the satellite imagines as described in section 1.10 and two different stratums were observed: For the region of Kaffrine maximum estimated shrub crown cover was 20 % (Stratum 2) and for the other areas the maximum shrub crown cover was 10 % (Stratum 1).[25] As the project area is subjected to periodic slash-and-burn practices, a crown cover, $CC_{SHRUB,i,t}$, equal to a one half of the observed maximum shrub crown covers is applied for each stratum.

[25] KML_Baseline Shrub Stratum1 and KML_Baseline_Shrub_Stratum2

	(b) Ex ante estimation of shrub crown cover at a time other than at the start of the project is carried out with the following considerations in view: (i) Shrub crown cover is assumed to remain at the preproject level unless transparent and verifiable information can be provided to justify a different rate of change; (ii) When land is abandoned, shrubs may encroach such land and shrub crown cover may reach the maximum value of 1.0 over a period of 20 years from the year in which the land is abandoned. If the year in which the land is abandoned is not known, then an average crown cover of 0.50 is assumed at the start of the project.

Data Unit / Parameter:	T
Data unit:	Year
Description:	Time period elapsed between two successive estimations of carbon stock in trees and shrubs
Source of data:	Recorded time
Description of measurement methods and procedures to be applied:	N/A
Frequency of monitoring/recording:	N/A
Value monitored:	N/A
Monitoring equipment:	N/A
QA/QC procedures to be applied:	N/A
Calculation method:	N/A
Any comment:	If the two successive estimations of carbon stock in trees are carried out at different points of time in year t_2 and t_1, (e.g. in the month of April in year t_1 and in the month of September in year t_2), then a fractional value is assigned to T.

<table>
<tr><td>Data Unit / Parameter:</td><td>$A_{BURN,i,t}$</td></tr>
<tr><td>Data unit:</td><td>Ha</td></tr>
<tr><td>Description:</td><td>Area burnt in stratum i in year t; ha</td></tr>
<tr><td>Source of data:</td><td>Field measurement or remote sensing measurement</td></tr>
<tr><td>Description of measurement methods and procedures to be applied:</td><td>The area shall be delineated either on the ground using GPS or from georeferenced remote sensing data</td></tr>
<tr><td>Frequency of monitoring/recording:</td><td>This area is measured whenever forest fire has occurred</td></tr>
<tr><td>Value monitored:</td><td>For the first project instance, for the first monitoring period:
<table>
<tr><td>Year</td><td>$A_{BURN,i,t}$ (ha)</td></tr>
<tr><td>03/06/2009-02/06/2010</td><td>0</td></tr>
<tr><td>03/06/2010-02/06/2011</td><td>0</td></tr>
<tr><td>03/06/2011-02/06/2012</td><td>0</td></tr>
<tr><td>03/06/2012-02/06/2013</td><td>17.80[26]</td></tr>
</table></td></tr>
<tr><td>Monitoring equipment:</td><td>GPS</td></tr>
<tr><td>QA/QC procedures to be applied:</td><td>Quality control/quality assurance (QA/QC) procedures prescribed under national forest inventory are applied. In the absence of these, QA/QC procedures from published handbooks, or from the IPCC GPG LULUCF 2003, may be applied</td></tr>
<tr><td>Calculation method:</td><td>N/A</td></tr>
<tr><td>Any comment:</td><td>Non-CO_2 GHG emissions resulting from any occurrence of fire within the project boundary shall be accounted for each incidence of fire which affects an area greater than the minimum threshold area reported by the host Party for the purpose of defining forest (i.e. 0.5 ha), provided that the accumulated area affected by such fires in a given year is ≥5% of the project area.[27]

17.80 ha > 0.5 ha (i.e the minimum threshold area of forest reported by Senegal) but only 3 % of the project area of the first project instance</td></tr>
</table>

[26] KML_Stratum8
[27] EB 65, Annex 31. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-08-v4.0.0.pdf

	and, therefore, non-CO_2 GHG emissions resulting from occurrence of this fire is not accounted for in accordance of the tool "Estimation of non-CO_2 greenhouse gas (GHG) emissions resulting from burning of biomass attributable to an A/R CDM project activity" [28]

Data Unit / Parameter:	$A_{p,i}$
Data unit:	ha
Description:	Area of sample p in stratum i
Source of data:	Field observation
Description of measurement methods and procedures to be applied:	Standard operating procedures (SOPs) prescribed under national forest inventory are applied. In the absence of these, SOPs from published handbooks, or from the IPCC GPG LULUCF 2003, are applied
Frequency of monitoring/recording:	Every five years since the year of the initial verification
Value monitored:	For the first project instance: <table><tr><th>Stratum</th><th>A_{p,i} (ha)</th></tr><tr><td>Stratum 1</td><td>0.04</td></tr><tr><td>Stratum 2</td><td>0.02</td></tr><tr><td>Stratum 3</td><td>0.00</td></tr><tr><td>Stratum 4</td><td>0.16</td></tr><tr><td>Stratum 5</td><td>0.06</td></tr><tr><td>Stratum 6</td><td>0.10</td></tr><tr><td>Stratum 7</td><td>0.07</td></tr><tr><td>Stratum 8</td><td>0.00</td></tr></table>
Monitoring equipment:	GPS
QA/QC procedures to be applied:	Quality control/quality assurance (QA/QC) procedures prescribed under national forest inventory are applied. In the absence of these, QA/QC procedures from published handbooks, or from the IPCC GPG LULUCF 2003, are applied
Calculation method:	The real area of sample will be confirmed at the time of each verification to correspond the the area determined as follows: The area of each

[28] EB 65, Annex 31. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-08-v4.0.0.pdf

	sample plot is 0.01 ha (50m * 2m). The total number of sample plots in each stratum is calculated with “Calculation of the number of sample plots for measurements within A/R CDM project activities” (version 02.1.0)[29].
Any comment:	Sample plot location is registered with a GPS and marked on the project map[30]

Data Unit / Parameter:	n_i
Data unit:	Dimensionless
Description:	Number of sample plots in stratum i
Source of data:	Calculated
Description of measurement methods and procedures to be applied:	N/A
Frequency of monitoring/recording:	n_i is calculated for each monitoring event, at least every five years
Value monitored:	For the first project instance: **Stratum** / $\mathbf{n_i}$: Stratum 1: 4 Stratum 2: 2 Stratum 3: 0 Stratum 4: 16 Stratum 5: 6 Stratum 6: 10 Stratum 7: 7 Stratum 8: 0
Monitoring equipment:	N/A
QA/QC procedures to be applied:	N/A
Calculation method:	The calculation method is described in the tool “Calculation of the number of sample plots for measurements within A/R CDM project activities” (version 02.1.0)[31]
Any comment:	See Excel-file: Sample plot size[32]

[29] EB 58, Annex 15. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-03-v2.1.0.pdf
[30] KML_Monitoring_Sample_plots
[31] EB 58, Annex 15. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-03-v2.1.0.pdf
[32] See Excel-file “Sample_plot_size”

Data Unit / Parameter:	w_i
Data unit:	Dimensionless
Description:	Ratio of the area of stratum i to the sum of areas of biomass estimation strata
Source of data:	Calculated
Description of measurement methods and procedures to be applied:	N/A
Frequency of monitoring/recording:	w_i is calculated for each monitoring event, at least every five years
Value monitored:	For the first project instance, first monitoring period: **Stratum** / **w_i** Stratum 1: 0.05 Stratum 2: 0.03 Stratum 3: 0.00 Stratum 4: 0.26 Stratum 5: 0.14 Stratum 6: 0.28 Stratum 7: 0.24 Stratum 8: 0.00
Monitoring equipment:	N/A
QA/QC procedures to be applied:	N/A
Calculation method:	w_i is equal to the area of the stratum i divided by the project area
Any comment:	See Excel-file: Sample plot size[33]

Data Unit / Parameter:	B_D
Data unit:	mm
Description:	Diameter at stem base
Source of data:	Physical measurements
Description of measurement methods and procedures to be applied:	Physical measurement of the trees in the sample plots, measurement taken at stem base.
Frequency of monitoring/recording:	B_D of trees in sample plots is measured and recorded every five years by ANOC

[33] See Excel-file “Sample_plot_size”

Value monitored:	See the attached Excel-file called "Monitoring sampling results"[34]
Monitoring equipment:	Measuring tape, paper template, computer
QA/QC procedures to be applied:	Part of overall QA/QC procedures discussed in Section 4.3 described in the project description document:10 % of the sample plots (5 sample plots) have been re-measured.
Calculation method:	N/A
Any comment:	N/A

Data Unit / Parameter:	N
Data unit:	Dimensionless
Description:	Total number of possible sample plots within the project boundary (the sampling space or the population)
Source of data:	Calculated
Description of measurement methods and procedures to be applied:	N/A
Frequency of monitoring/recording:	N is calculated for each monitoring event, at least every five years.
Value monitored:	For the first project instance: 57.133
Monitoring equipment:	N/A
QA/QC procedures to be applied:	N/A
Calculation method:	N is equal to project area of the first project instance divided by the size of the sample plot (571.33 ha/0.01 ha = 57.133)
Any comment:	See Excel-file: Sample plot size[35]

Data Unit / Parameter:	s_i
Data unit:	t d.m. ha^{-1}
Description:	Estimated standard deviation of biomass stock in stratum i. Standard deviation of biomass stock per unit area (in t d.m. ha^{-1}) may also be used for this purpose.
Source of data:	Calculated

[34] See Excel-file "Monitoring sampling results"
[35] See Excel-file "Sample_plot_size"

Description of measurement methods and procedures to be applied:	Approximate value of the standard deviation of biomass stock in each stratum is either known from existing data related to the project area or existing data related to a similar area, or is estimated from a preliminary sample
Frequency of monitoring/recording:	s_i is calculated for each monitoring event, at least every five years.
Value monitored:	For the first project instance: **Stratum** / **s_i** Stratum 1: 21.1 Stratum 2: 14.2 Stratum 3: 9.5 Stratum 4: 13.4 Stratum 5: 9.0 Stratum 6: 8.5 Stratum 7: 6.4 Stratum 8: 7.5
Monitoring equipment:	N/A
QA/QC procedures to be applied:	N/A
Calculation method:	50 % of the estimated mean biomass stock
Any comment:	See Excel-file: Sample plot size[36]

Data Unit / Parameter:	A_i
Data unit:	ha
Description:	The area of stratum i of the areas of land; ha
Source of data:	Field measurement
Description of measurement methods and procedures to be applied:	The area shall be delineated either on the ground using GPS or from georeferenced remote sensing data
Frequency of monitoring/recording:	Every five years since the year of the initial verification
Value monitored:	N/A
Monitoring equipment:	GPS
QA/QC procedures to be applied:	Quality control/quality assurance (QA/QC) procedures prescribed under national forest inventory are applied. In the absence of these,

[36] See Excel-file "Sample_plot_size"

	QA/QC procedures from published handbooks, or from the IPCC GPG LULUCF 2003, are applied
Calculation method:	N/A
Any comment:	A_i is used for calculating the change in SOC stock. SOC it is not accounted for the first project instance.

3.3 Description of the Monitoring Plan

The monitoring plan of the project activity meets the requirements established by the applied methodology AR-ACM0003 (Version 01.0.0). In accordance with the applied methodology the monitoring plan shall provide for collection of all relevant data necessary for

(a) Verification that the applicability conditions listed under paragraphs 3 and 4 have been met;

(b) Verification of changes in carbon stocks in the pools selected; and

(c) Verification of project emissions and leakage emissions.

The data collected will be archived for a period of at least two years after the end of the last crediting period of the project activity. The precision requirements are those listed in the Verified Carbon Standard and in the tool "Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities"[37].

The organizational structure of monitoring is presented below in Figure 3.

[37] AR-TOOL14. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-14-v3.0.0.pdf

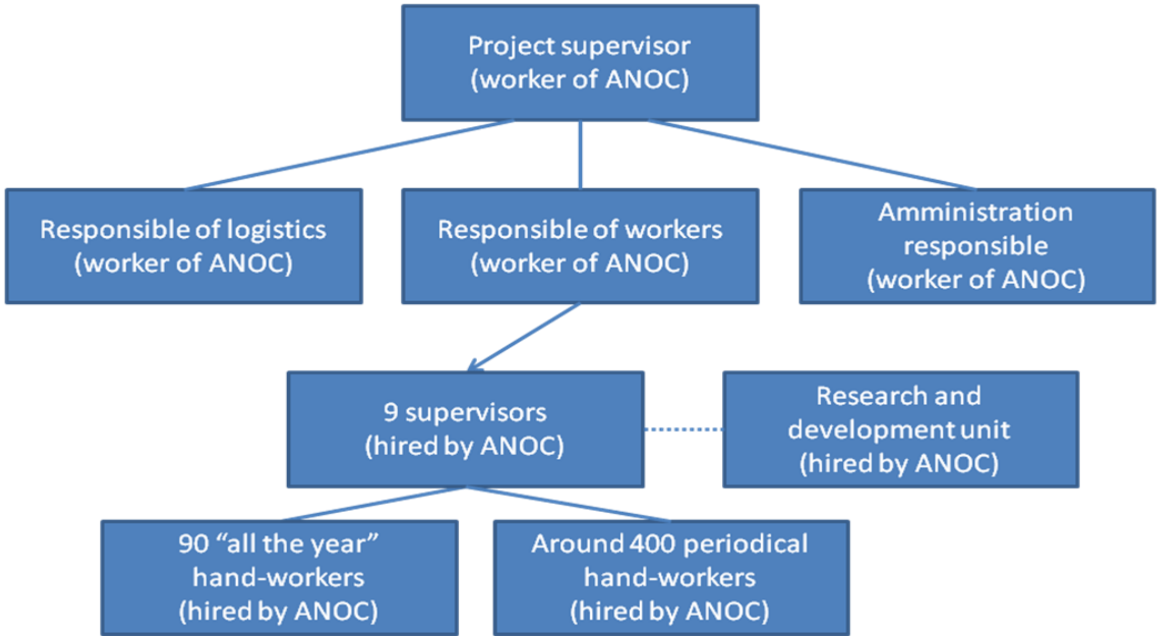

Figure 3. Organization structure of the project management

a) Verification that the applicability conditions listed under paragraphs 3 and 4 have been met

The verification of the applicability conditions is done through the project boundary monitoring and through forest establishment monitoring.

Monitoring of project boundary has been conducted with field surveys concerning the project boundary within which the project activity has occurred, site by site, measuring geographical positions using GPS and checking that the planted areas are in coherence with the eligibility criteria. Geographic boundaries of the project area has been recorded in a database and archived. Personnel involved in the monitoring have been trained to identify the changes in the boundary and to record changes in the project database for reporting of project verification. For the first monitoring period the following findings have been made about the project boundary:

- The project boundaries of the first project instance outlined in the project plan (VSC-PD) were confirmed to hold for the *ex post* situation.[38,39]
- All the areas of the first project instance are in coherence with the eligibility criteria described in section 1.13 of the VCS-PD.
- The forest planting and management hass been done in accordance with the original plans provided in the VCS-PD.

[38] KML_Plantations_2009-2010-2011
[39] Google_Earth_imagines_for_baseline_and_current_situation

Monitoring of forest establishment and forest management. The inventory operations, including field data collection and data management, has been done following standard operating procedures (SOPs) and quality control/quality assurance (QA/QC) procedures. Established stands have been monitored with respect to the species and strata pre-defined in the VCS-PD. During the first monitoring period there have not been documented deviations from the planned forest establishment.

The survival of the samplings has been quantified in the field approximately six months after the planting of each plot. The survival check has been repeated every 6th month during the first three years after the planting. If the survival was below 95% of the initial quantity planted, the stand has been replanted with the same species, seeking to maintain the plots homogeneous in terms of age and development. The estimate is made through a simple count of the individuals within each plot, taking on account their vitality state, determining the density of live individuals and finally comparing it to the initial quantity. During the first monitoring period It has been noted that keeping the homogeneity (in terms of age and development) of the plantations is challenging and the replanted seedlings have been in most cases younger than the original plantations of that plot.

b) Verification of changes in carbon stocks in the pools selected

<u>Forest management activities</u> (e.g., planting, re-planting, pruning and fruit harvesting) as well as <u>unexpected natural disturbances</u> occurring during the crediting period (e.g., due to fire, pests or disease outbreaks) has been monitored to guarantee that the correct practices are applied in accordance with the management plan and to ensure the health of the stands. In case of the wild fires occurs the areas subjected to the fire are monitored in accordance with the tool "Estimation of non-CO_2 GHG emissions resulting from burning of biomass attributable to an A/R CDM project activity[40].

During the monitoring period there has not been any extensive outbreaks caused by pests or diseases which would have required replanting. There has anyhow occurred one wild fire which destroyed the plantations in the plot Dianke Kao (17.66 ha)[41] in the end of November 2012. This plot has been replanted in 2013 respecting the originally selected planting density (1111 trees/ha).

<u>Above and below ground biomass</u>

Pre-existing (baseline) trees and non-tree vegetation have not been measured and accounted for, making the monitoring conservative. Anyhow, the monitoring has included verifying that the baseline trees were not removed i.e monitoring the parameter $CC_{TREE\_BSL,i}$. Ocular estimations of the pre-project trees has been made during the monitoring period and it was confirmed that the *ex ante* estimation, $CC_{TREE\_BSL,i} = 0.01$, still holds on. Pre-existing and planted trees could be easily differentiated because pre-existing species are different to the planted (Jatropha curcas L.). The not removing the baseline trees has been also confirmed

[40] EB 65, Annex 31. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-08-v4.0.0.pdf

[41] KML_Stratum8

by comparing the satellite imagines taken before project start date with resent satellite imagines taken from the project area[42].

The growth of the Jatropha trees has been monitored by measuring the diameters at stem base of each tree located in the sample plots following a detailed pre-described procedure[43]. The results have been used for the estimation of above-ground tree biomass using allometric model based on stem base diameter. The below ground biomass have been extrapolated from above-ground biomass.

Sample plots establishment and trees measurements has been done using the following equipments: Measuring tape, measuring cord, GPS and compass[44]. All the measurements results has been recorded on the field notes[45].

Figure 4. Determining the sample plot location using the GPS, compass and the 50 meters long measurement cord. (Author: ANOC)

[42] Google_Earth_imagines_for_baseline_and_current_situation
[43] Rilevamento_parametri_biometrici, CarbonSinkGroup 2013
[44] Photos_about the monitoring sampling
[45] Monitoring_field_notes_1 and Monitoring_field_notes_2

Picture 5. Measuring the stem base circumference and recording the results on the field notes. (Author: ANOC)

The appropriateness of the selected allometric equation has been demonstrated in accordance with the tool "Demonstrating appropriateness of allometric equations for estimation of aboveground tree biomass in A/R CDM project activities" [46] as described in the document "Demonstrating appropriateness of allometric equation for the project "Jatropha Agroforestry Senegal"[47].

$$Fj\,(B_D) = 1.213 * 0.000907 * B_D^{3.354} \qquad \text{(equation 1)}$$

Where:

$fj(B_D)$ Total above ground biomass per tree (g d.m.)

B_D Basal diameter at stem base (mm)

Soil organic carbon,

The soil organic carbon (SOC) is not accounted for the first project instance as the litter has been removed from the project plots during the first project years.

deadwood and litter

The carbon pools in dead wood and in litter have not been selected to be accounted for as described in the VCS-PD (Section 2.3) and, therefore, they are not monitored.

[46] A/R Methodological tool number 17. Available at http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-17-v1.pdf

[47] Demonstrating appropriateness of allometric equation for the project "Jatropha Agroforestry Senegal".CarbonSinkGroup, 2013.

c) Verification of project emissions and leakage emissions

According the applied methodology the only increase in GHG emissions within the project boundary which needs to be accounted for is the non-CO_2 GHG emissions from burning of woody biomass attributable to an A/R CDM project activity. The monitoring of emissions is required only if the emissions are considered significant; if insignificant, evidence should be provided that the assumption for the exclusion made in the *ex ante* assessment still hold in the *ex post* situation.

The *ex ante* decision to account the increase in non-CO_2 GHG emissions within the project boundary as a result of the implementation of the A/R project activity ($GHG_{E,t}$) as zero was made because the project implementation plan was not foreseen fire use in site preparation or in clearing the land of harvest residue prior replanting of the land. Moreover the impact of wild fires was estimated to be insignificant[48]. All of these assumptions were confirmed to hold on during the first monitoring period:

- The eligibility criteria "Fire is not used in site preparation and/or to clear the land of harvest residue prior replanting of the land" has been fulfilled for the first project instance during the first monitoring period.
- During the monitoring period, there has recorded to have occurred one wildfire, in year 2012. The burned area has been monitored in accordance with the tool "Estimation of non-CO_2 GHG emissions resulting from burning of biomass attributable to an A/R CDM project activity[49]" to be less 17.66 ha which is less than 5 % of the total project area of the first project instance. Therefore, in accordance of the above mentioned tool, the increase in non-CO_2 GHG emissions can be considered insignificant.

Under applicability conditions of the applied methodology the only leakage emissions that can occur are the GHG emissions due to displacement of pre-project activities. As described in section 2.4 of the project design document for the *ex ante* situation the leakage emissions has been estimated to be insignificant. Resent satellite imagines[50] can confirm that the project area or it's the adjacent areas does not contain any significant crop cultivation or grazing activities which would have been displaced because of the project activity. The assumptions made *ex ante* is therefore confirmed to hold on still in the *ex post* situation and thus the leakage can be estimated to be zero also for the *ex post* situation.

[48] Non-CO_2 GHG emissions resulting from any occurrence of fire within the project boundary shall be accounted for each incidence of fire which affects an area greater than the minimum threshold area reported by the host Party for the purpose of defining forest, provided that the accumulated area affected by such fires in a given year is ≥5% of the project area. In other cases the impact of the fires can be estimated insignificant.
[49] EB 65, Annex 31. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-08-v4.0.0.pdf
[50] See the attached document "Google_Earth_imagines_for_baseline_and_current_situation"

d) Stratification

Stratification of project area was done *ex ante* based on the project planting/management plan. The *ex ante* stratification was updated *ex post* because during the first monitoring period there has occured a wild fire inside the project boundaries which have been estimated to have added variability to the growth pattern of the biomass in the project area. The list of plantation area inside each defined stratum is presented in a separate Excel-file[51] and the borders of each stratum in separate KML-files[52].

Table 2 shows the eight *ex post* strata defined for project activities based on the plantation age and the planting density of the trees. In the future, during the crediting period, the changes of project strata will be monitored adjusting the sampling frame to the changes in the number and extension of strata.

Table 2. *Ex post* stratification

***Ex post* strata**	**Planting year**	**Planting density (trees/ha)**	**Area (ha)**
Stratum 1	2009	2,500	29.36
Stratum 2	2009	1,666	18.66
Stratum 3	2009	1,111	1.11
Stratum 4	2010	1,666	144.11
Stratum 5	2010	1,111	76.40
Stratum 6	2011	1,111	154.05
Stratum 7	2012	833	129.84
Stratum 8	2013*	1,111	17.80

*Originally planted in 2010. Replanted in 2013 after occurrence of fire.

e) Sampling framework

Sample size

Permanent sampling plots are used for sampling over time to measure and monitor changes in carbon stocks. Each sampling plot is 50 meter long and 2 meter wide row (0.01 ha). This size and shape is chosen to guarantee that there will be approximately at least 10 trees within the plot boundaries at the end of the crediting period; The smallest initial density of plantation will be 3x4 m (see table 2) and as the recovery rate of plantations is assumed to be around 95%.

The total number of permanent sampling pots was estimated with the tool “Calculation of the number of sample plots for measurements within A/R CDM project activities” (version 02.1.0)[53]. Sample plots have been located randomly inside each stratum and measured for verification purposes.

[51] See Excel-file “Stratification_ex_post”
[52] See KML-files for ex post stratums 1-8
[53] EB 58, Annex 15. Calculation of the number of sample plots for measurement within A/R CDM project activities (Version 02.1.0). http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-03-v2.1.0.pdf

The total number of sample plots was determined to be 45 (Table 3). The number was calculated with the below presented equation 2 and then added 5% more plots as additional quality guarantee.[54]

$$n = \frac{N \cdot t_{VAL}^{\ 2} \cdot \left(\sum_i w_i \cdot s_i \right)^2}{N \cdot E^2 + t_{VAL}^{\ 2} \cdot \sum_i w_i \cdot s_i^{\ 2}} \qquad \text{(equation 2)}$$

Where:

Parameter	Description	Comment
n	Number of sample plots required for estimation of biomass stocks within the project the project boundary; dimensionless	Calculated with equation 2
N	Total number of possible sample plots within the project the project boundary (i.e., the sampling space or the population); dimensionless	57.133which is equal to the project area (751.33 ha) divided by the size of the sample plot (0.01 ha)
t_{VAL}	Two-sided Student´s t-value, at infinite degrees of freedom, for the required confidence level; dimensionless	1.960 as presented in the used tool (Confidence level of 95%, degree of freedom of "infinite)
w_i	Relative weight of the area of stratum *i* (i.e., the area of the stratum *i* divided by the project area); dimensionless	The relative weight of the area of stratum *i* is equal to the area of the stratum *i* divided by the project area
s_i	Estimated standard deviation of biomass stock in stratum *i*; t.d. (t d.m. ha^{-1})	A standard deviation of 50% of the mean biomass stock within the stratum is used [55]
E	Acceptable margin of error (i.e., one-half the confidence interval) in estimation of biomass stock within the project boundary (i.e., in the units used for s_i); t d.m. (or t d.m. ha^{-1})	A default value equal to 15% of the mean biomass stock within the project is used
i	*1,2,3*... Biomass stock estimation strata within the project boundary	See the table 2

[54] See the Excel-file "Sample_plot_size"
[55] Paragraph 6 of "Guidelines on conservative choice and application of default data in estimation of the net anthropogenic GHG removals by sinks" (version 02): http://cdm.unfccc.int/Reference/Guidclarif/ar/methAR_guid26.pdf

Plots location

The allocation of sample plots among the strata was estimated by following the tool “Calculation of the number of sample plots for measurements within A/R CDM project activities” (version 02.1.0)[56] using the following equation (equation 3):

$$n_i = n \cdot \frac{w_i \cdot s_i}{\sum_i w_i \cdot s_i} \qquad \text{(equation 3)}$$

Where:

Parameter	Description	Comment
n_i	Number of sample plots allocated to stratum *i*, dimensionless	Calculated with equation 3
n	Number of sample plots required for estimation of biomass stocks within the project boundary; dimensionless	Calculated with equation 2
w_i	Relative weight of the area of stratum *i* (i.e., the area of the stratum *i* divided by the project area); dimensionless	The relative weight of the area of stratum *i* is equal to the area of the stratum *i* divided by the project area
s_i	Estimated standard deviation of biomass stock in stratum *i*; t.d. (t d.m. ha^{-1})	A standard deviation of 50% of the mean biomass stock within the stratum is used [57]
i	*1,2,3*... Biomass stock estimation strata within the project boundary	See the table 2

[56] EB 58, Annex 15. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-03-v2.1.0.pdf
[57] Paragraph 6 of “Guidelines on conservative choice and application of default data in estimation of the net anthropogenic GHG removals by sinks” (version 02). http://cdm.unfccc.int/Reference/Guidclarif/ar/methAR_guid26.pdf

Table 3. Number and allocation of the sample plots for the *ex post* stratums

Stratum	Number of sample plots allocated to stratum *i* (n_i)
Stratum 1	4
Stratum 2	2
Stratum 3	0
Stratum 4	16
Stratum 5	6
Stratum 6	10
Stratum 7	7
Stratum 8	0*
Total number of sample plots (n)	45

* Not any sample was selected for the Stratum 8 as the plantations of this Stratum were recently replanted after the occurrence of a wild fire. The Stratum 8 is not included in the ex post biomass estimations of the first verification event i.e. the biomass for this Stratum is conservatively accounted as zero for the first monitoring event.

Permanent sample plots were located randomly inside each stratum. The random location was determined with a step wise procedure: First, the allocation inside the different districts of each stratum was identified randomly. After the random districts had been selected, the random geographical coordinates inside the selected districts were created[58]. The coordinates were used as a reference for the starting point of each individual permanent sample plot. The concrete sampling procedure is described more detailed in separate document “Rilevamento parametri biometrici”[59] and the calculation procedures in the file “Sample_plot_size”.[60]

Selected sample plots were identified on site using GPS and marked on maps to facilitate their location during future field work[61]. For each monitoring period, maps will be updated and the data on new planted areas and eventual new sampling lots will be included. Plots has not be marked on site to ensure that they do not receive differential treatment.

f) Monitoring frequency

Although the verification and certification will be carried out every five years after the first verification until the end of the crediting period, the monitoring interval can be less than five years so that enough information can be collected for management purposes. However, it is possible that only the data to be used for verification will be recorded in the database structure used for verifications. Additional data may be kept in a separate bookkeeping system.

[58] Random Point Generator of Geo Midpoint is used for creating the coordinates. http://www.geomidpoint.com/random/
[59] Rilevamento parametri biometrici, CarbonSinkGroup 2013
[60] See the Excel-file “Sample_plot_size”
[61] KML_Monitoring_Sample_plots

g) Quality Assurance/Quality Control plan

Quality assurance

A quality assurance/quality control (QA/QC) plan has been implemented to ensure the net anthropogenic GHG removals by sinks are measured and monitored precisely, credibly, verifiably and transparently. QA/QC control includes steps to control the errors in project boundary, stratification, sampling, measurement, data entry, data analysis, and data maintenance and archiving. During the first monitoring period this plan was implemented in two phases, as follows:

First phase: The objective of this phase was to train the measurement group. The group consist of local people whose work is supervised by the ANOC workers (see Figure 3). The training included teaching the group leaders to manage and calibrate the various measuring instruments and to understand and analyze their use. The models and parameters needed for calculating were also explained and analyzed in the first phase.

The first phase included also specifics training for local farmers in Jatropha cultivation, sustainable agricultural and forest management practices. A comprehensive illustrated book on agricultural and forest management practices has been produced, printed and distributed among farmers and representatives in schools and villages.

Second phase: All the theoretical knowledge acquired in the first phase was put into practice in this phase. The phase included a practice of the use the instruments. The following activities were conducted to confirm the proper use of the instruments:

- With the compass and GPS, the groups assigned to each area established and measured permanent plots, and calculated their area.
- Once plots were established, information on the trees found inside them was collected, using appropriated instruments and paper forms[62]. To check the accuracy of the information, the specialist from ANOC conducted a separate measurements on five randomly selected sample plots which included totally 75 trees[63] and the results were compared with those obtained by the trained groups and the quality of the made measurements were confirmed as the measurement results obtained by the trained groups and the control measurements were similar approximately in 95 % of the cases. [64]

Quality control

ANOC is responsible for planning and for designing a system to control the quality of growth measurements. ANOC has collect samples which are used to make comparisons with the information collected by the teams assigned to each area. This quality control activity has been conducted ones after the original field measured. Meetings has been held with the

[62] Monitoring_field_notes_1
[63] Monitoring_field_notes_2
[64] See Excel-file “Monitoring sampling results”

persons responsible for processing information in order to review formats, the screening system for information collected, and the databases and calculations.

Survey data has been entered into a computer-based information system especially designed for the project. This makes it possible to record, calculate and analyze all inventory information, including calculations of the biomass of permanent parcels. To minimize the possible errors in the process of data entry, the entered data has been reviewed by an independent expert and, where necessary, compared with independent data to ensure that the data are realistic. Communication between all personnel involved in measuring and analyzing of data will be used to resolve any apparent anomalies before the final analysis of the monitoring data. If there are any problems with the monitoring plot data that cannot be resolved, the plot is not used in the analysis. All the electronic data and reports has been copied on durable media, such as compact discs (CDs), and copies of them are stored in multiple locations.

Conservative approach and uncertainties

The guidelines provided in the document "Guidelines on conservative choice and application of default data in estimation of the net anthropogenic GHG removals by sinks" (Version 02)[65] is used to ensure that application of default data in estimation of the net anthropogenic GHG removals by sinks results in conservative, but not overly conservative, estimates.

h) Record system

The project database includes all information related to the monitoring of project activities: identification codes and coordinates for each sampling plot, dates when sampling has been made, persons involved in the sampling and the sampling results. Database will include also scanned original field notes.

The database ensures that all afforested and/or reforested areas within a specific project instance are uniquely defined and are included exclusively in one project, thereby avoiding double accounting of emission removals. The project participant (ANOC) have in a concession all the lands where the project activities are implemented. The procedure to ensure that double accounting does not occur is the confirmation that a new project instances are not included in the above mentioned project database or in the CDM's project activities database (UNFCCC).

[65] EB 50, Annex 23. Guidelines on conservative choice and application of default data in estimation of the net anthropogenic GHG removals by sinks. http://cdm.unfccc.int/Reference/Guidclarif/ar/methAR_guid26.pdf

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

Procedures to be used for calculation of baseline net GHG removals by sinks are detailed in the AR-ACM0003 methodology (Version 01.0.0) under the section 5.4 “Baseline net GHG removals by sinks”. According the methodology baseline net GHG removals by sinks are calculated with the following equation:

$$\Delta C_{BSL,t} = \Delta C_{TREE\_BSL,t} + \Delta C_{SHRUB\_BSL,t} + \Delta C_{DW\_BSL,t} + \Delta C_{LI\_BSL,t} \qquad \text{(equation 4)}$$

Where:

$\Delta C_{BSL,t}$	Baseline net GHG removals by sinks in year t; tCO_2-e
$\Delta C_{TREE\_BSL,t}$	Change in carbon stock in baseline tree biomass within the project boundary in year t, as estimated in the tool “Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities[66]”; tCO_2-e
$\Delta C_{SHRUB\_BSL,t}$	Change in carbon stock in baseline shrub biomass within the project boundary, in year t, as estimated in the tool “Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities[67]”; tCO_2-e
$\Delta C_{DW\_BSL,t}$	Change in carbon stock in baseline dead-wood biomass within the project boundary, in year t, as estimated in the tool “Estimation of carbon stocks and change in carbon stocks in dead wood and litter in A/R CDM project activities[68]”; tCO_2-e
$\Delta C_{LI\_BSL,t}$	Change in carbon stock in baseline litter biomass within the project boundary, in year t, as estimated in the tool “Estimation of carbon stocks and change in carbon stocks in dead wood and litter in A/R CDM project activities[69]”; tCO_2-e

Carbon stock in trees

a) According the tool “Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities[70] change in carbon stock in trees in the baseline is estimated using the following stepwise procedure: Select the technique and use the appropriate equation for estimating the carbon stock in trees.

Here is chosen “Baseline default technique”

$$C_{TREE,BSL,i} = \frac{44}{12} * CF_{TREE\_BSL} * B_{FOREST} * (1 + R_{TREE\_BSL}) * CC_{TREE\_BSL,i} * A_{BSL,i} \qquad \text{(equation 5)}$$

[66] AR-TOOL14. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-14-v3.0.0.pdf
[67] AR-TOOL14. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-14-v3.0.0.pdf
[68] EB 67, Annex 23. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-12-v2.0.0.pdf
[69] EB 67, Annex 23. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-12-v2.0.0.pdf
[70] AR-TOOL14. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-14-v3.0.0.pdf

$$C_{TREE,BSL} = \sum_{i=1}^{M} C_{TREE\_BSL,i} \qquad \text{(equation 6)}$$

Where:

$C_{TREE,BSL}$	Carbon stock in living trees in the baseline, in the project boundary, at the start of the A/R project activity; t CO2-e
$C_{TREE,BSL,i}$	Carbon stock in living trees in the baseline, in baseline stratum i, at the start of the A/R project activity; t CO2-e. Baseline strata are delineated on the basis of tree crown cover
CF_{TREE_BSL}	Carbon fraction of tree biomass in the baseline; t C $(t.d.m.)^{-1}$ A default value of 0.47 t C $(t.d.m.)^{-1}$ is used
B_{FOREST}	Default above-ground biomass content in forest in the region/country where the A/R project is located; t d.m. ha^{-1}
R_{TREE_BSL}	Root-shoot ratio for the trees in the baseline; dimensionless. A default value of 0.25 is used unless transparent and verifiable information can be provided to justify a different value
$CC_{TREE\_BSL,i}$	Crown cover of trees in the baseline, in baseline stratum i, at the start of the A/R project activity, expressed as a fraction (e.g. 10% crown cover implies $CC_{TREE\_BSL,i} = 0.10$); dimensionless
$A_{BSL,i}$	Area of stratum i in the baseline, delineated on the basis of tree crown cover at the start of the A/R project activity; ha
i	1, 2, 3, … tree biomass estimation strata within the project boundary

b) Select the method and use the appropriate equation for estimating the carbon stock change in trees.

Here is chosen “Baseline default method” which is acceptable method in accordance of the used tool as the mean tree crown cover in the baseline is less than 6 % (i.e. less than 20 per cent of the threshold crown cover reported by Senegal, 30 %).

$$\Delta C_{TREE,BSL,i} = \frac{44}{12} * CF_{TREE\_BSL} * \Delta B_{FOREST} * (1 + R_{TREE\_BSL}) * CC_{TREE\_BSL,i} * A_{BSL,i} \qquad \text{(equation 7)}$$

$$\Delta C_{TREE,BSL} = \sum_{i=1}^{M} \Delta C_{TREE\_BSL,i} \qquad \text{(equation 8)}$$

Where:

$\Delta C_{TREE,BSL}$	Average annual change in carbon stock in tree biomass in the baseline; t CO2-e yr^{-1}
$\Delta C_{TREE,BSL,i}$	Average annual change in carbon stock in tree biomass in the baseline in baseline stratum i; t CO2-e yr-1
CF_{TREE_BSL}	Carbon fraction of tree biomass in the baseline; t C $(t.d.m.)^{-1}$ A default value of 0.47 t C $(t.d.m.)^{-1}$ is used
ΔB_{FOREST}	Default average annual increment of above-ground biomass in forest in the region/country where the A/R project is located; t d.m. ha^{-1} yr^{-1}
R_{TREE_BSL}	Root-shoot ratio for the trees in the baseline; dimensionless. A default value of 0.25 is used unless transparent and verifiable information can be provided to justify a different value
$CC_{TREE\_BSL,i}$	Crown cover of trees in the baseline, in baseline stratum i, at the start of the A/R project activity, expressed as a fraction (e.g. 10% crown cover implies $CC_{TREE\_BSL,i} = 0.10$); dimensionless
$A_{BSL,i}$	Area of stratum i in the baseline, delineated on the basis of tree crown cover at the start of the A/R project activity; ha
i	1, 2, 3, … tree biomass estimation strata within the project boundary

Carbon stock in shrubs

According the tool “Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities[71] change in carbon stock in shrubs in the baseline is estimated:

$$C_{SHRUB,t} = \frac{44}{12} * CF_s * (1 - R_s) * \sum_i A_{SHRUB,i,t} * B_{SHRUB,i,t} \quad \text{(equation 9)}$$

For those areas where the shrub crown cover is less than 5 %, the shrub biomass per hectare is considered negligible and hence accounted as zero. Instead, for those areas where the shrub crown cover is 5 % or more, shrub biomass per hectare is estimated as follows:

$$B_{SHRUB,i,t} = BDR_{SF} \times B_{FOREST} \times CC_{SHRUB,i,t} \quad \text{(equation 10}$$

$$dC_{SHRUB,(t_1 t_2)} = \frac{C_{SHRUB,t_2} - C_{SHRUB,t_1}}{T} \quad \text{(equation 11)}$$

$$\Delta C_{SHRUB,t} = dC_{SHRUB,(t_1,t_2)} \times 1 year \;\; for\; t_1 \le t_2 \quad \text{(equation 12)}$$

[71] AR-TOOL14. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-14-v3.0.0.pdf

Where:

$C_{SHRUB,t}$	Carbon stock in shrub biomass within the project boundary at a given point of time in year t; t CO2-e
CF_s	Carbon fraction of shrub biomass; t C $(t.d.m.)^{-1}$ IPCC default value of 0.47 t C $(t.d.m.)^{-1}$ is used
R_s	Root-shoot ratio for shrubs; dimensionless
$A_{SHRUB,i,t}$	Area of shrub biomass stratum i at a given point of time in year t; ha
$B_{SHRUB,i,t}$	Shrub biomass per hectare in shrub biomass stratum i at a given point of time in year t; t d.m. ha^{-1}
BDR_{SF}	Ratio of shrub biomass per hectare in land having a shrub crown cover of 1.0 and default above-ground biomass content per hectare in forest in the region/country where the A/R project is located; dimensionless
B_{FOREST}	Default above-ground biomass content in forest in the region/country where the A/R project is located; t d.m. ha^{-1}
$CC_{SHRUB,i,t}$	Crown cover of shrubs in shrub biomass stratum i at a given point of time in year t expressed as a fraction (e.g. 10% crown cover implies $CC_{SHRUB,i,t} = 0.10$); dimensionless
$dC_{SHRUB,(t1,t2)}$	Rate of change in carbon stock in shrub biomass within the project boundary during the period between a point of time in year t_1 and a point of time in year t_2; t CO_2-e yr^{-1}
$C_{SHRUB,t2}$	Carbon stock in shrub biomass within the project boundary at a point of time in year t_2; t CO_2-e
$C_{SHRUB,t1}$	Carbon stock in shrub biomass within the project boundary at a point of time in year t_1; t CO_2-e
$\Delta C_{SHRUB,t}$	Change in carbon stock in shrub biomass within the project boundary in year t; t CO_2-e
T	Time elapsed between two successive estimations ($T=t_2 - t_1$); yr
i	1, 2, 3, … shrub biomass strata delineated on the basis of shrub crown cover
t	1, 2, 3, … years counted from the start of the A/R project activity

Carbon stocks in dead wood and litter

The carbon stock in dead wood and litter are not selected (See table 6 of the PD) and thus in accordance with the applied methodology these pools are set to zero.

4.2 Project Emissions

Procedure to be used for calculation of actual net GHG removals by sinks are detailed in the AR-ACM0003 methodology (Version 01.0.0) under the section 5.5 “Actual net GHG removals by sinks”. According the methodology the actual net GHG removals by sinks is calculated as follows:

$$\Delta C_{ACTUAL,t} = \Delta C_{P,t} - GHG_{E,t} \quad \text{(equation 13)}$$

Where:

$\Delta C_{ACTUAL,t}$	Actual net GHG removals by sinks, in year t; tCO_2-e
$\Delta C_{P,t}$	Change in the carbon stocks in project, occurring in the selected carbon pools, in year t; tCO_2-e
$GHG_{E,t}$	Increase in non-CO_2 GHG emissions within the project boundary as a result of the implementation of the A/R CDM project activity, in year t, as calculated in the tool “Estimation of non-CO_2 GHG emissions resulting from burning of biomass attributable to an A/R CDM project activity[72]”; tCO_2-e

As described in paragraph “c) Verification of project emissions and leakage emissions” of section 3.3. of this document the Project emissions can be considered insignificant for the first monitoring period. Therefore $GHG_{E,t} = 0$.

Change in the carbon stocks in project, occurring in the selected carbon pools, in year *t* is calculated as follows:

$$\Delta C_{P,t} = \Delta C_{TREE\_PROJ,t} + \Delta C_{SHRUB\_PROJ,t} + \Delta C_{DW\_PROJ,t} + \Delta C_{LI\_PROJ,t} + \Delta SOC_{AL,t} \quad \text{(equation 14)}$$

Where:

$\Delta C_{P,t}$	Change in the carbon stocks in project, occurring in the selected carbon pools, in year t; tCO_2-e
$\Delta C_{TREE\_PROJL,t}$	Change in carbon stock in tree biomass in project in year t, as estimated in the tool “Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities[73]”; tCO_2-e

[72] EB 65, Annex 31. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-08-v4.0.0.pdf
[73] AR-TOOL14. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-14-v3.0.0.pdf

$\Delta C_{SHRUB\_PROJL,t}$	Change in carbon stock in shrub biomass in project in year t, as estimated in the tool "Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities[74]"; tCO_2-e
$\Delta C_{DW\_PROJL,t}$	Change in carbon stock in dead-wood biomass in project in year t, as estimated in the tool "Estimation of carbon stocks and change in carbon stocks in dead wood and litter in A/R CDM project activities[75]"; tCO_2-e
$\Delta C_{LI\_PROJL,t}$	Change in carbon stock in litter biomass in project in year t, as estimated in the tool "Estimation of carbon stocks and change in carbon stocks in dead wood and litter in A/R CDM project activities[76]"; tCO_2-e
$\Delta SOC_{AL,t}$	Change in carbon stock in SOC in project, in year t, in areas of land meeting the applicability conditions of the tool "Tool for estimation of change in soil organic carbon stocks due to the implementation of A/R CDM project activities[77]", as estimated in the same tool; tCO_2-e

Change in carbon stock in tree biomass in project

Carbon stock in tree biomass is estimated on the basis of the tree biomass strata described in the table 2. Carbon stock in trees is estimated by applying Allometric equation technique and change in carbon in trees is estimated by applying Stock change method (equations 15-25 below).

$$B_{TREE,j,p,i,t} = f_{j(x1p,i,t,x2p,i,t,x3p,i,t)} * (1+R_j) \qquad \text{(equation 15)}$$

$$B_{TREE,p,i,t} = \sum_j B_{TREE,j,p,i,t} \qquad \text{(equation 16)}$$

$$b_{TREE,p,i,t} = \frac{B_{TREE,p,i,t}}{A_{p,i}} \qquad \text{(equation 17)}$$

$$b_{TREE,i,t} = \frac{\sum_{p=1}^{n_i} b_{TREE,p,i,t}}{n_i} \qquad \text{(equation 18)}$$

$$s_i^2 = \frac{n_i * \sum_{p=1}^{n_i} b_{TREE,p,i,t}^2 - \left(\sum_{p=1}^{n_1} b_{TREE,p,i,t}\right)^2}{n_i * (n_i - 1)} \qquad \text{(equation 19)}$$

[74] AR-TOOL14. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-14-v3.0.0.pdf
[75] EB 67, Annex 23. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-12-v2.0.0.pdf
[76] EB 67, Annex 23. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-12-v2.0.0.pdf
[77] EB 60, Annex 12. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-16-v1.1.0.pdf

$$b_{TREE,t} = \sum_{i=1}^{M} w_i * b_{TREE,i,t}$$ (equation 20)

$$s^2_{b_{TREE,t}} = \sum_{i=1}^{M} w_i^2 * \frac{s_i^2}{n_i}$$ (equation 21)

$$u_{b_{TREE,t}} = \frac{t_{VAL} * s_{bTREE,t}}{b_{TREE,t}}$$ (equation 22)

$$C_{TREE,t} = \frac{44}{12} \times B_{TREE,t} * CF_{TREE}$$ (equation 23)

$$dC_{TREE(t1,t2)} = \frac{C_{TREE,t2} - C_{TREE,t1}}{T}$$ (equation 24)

$$\Delta C_{TREE,t} = \Delta C_{TREE,(t1,t2)} * 1 year \text{ for } t_1 \leq t \leq t_2$$ (equation 25)

Where:

$B_{TREE,i,p,i,t}$	Biomass of trees of species j in sample plot p of stratum i at a given point of time in year t; t d.m.
$f_{j(x1p,i,t,x2p,i,t,x3p,i,t)}$	Function relating measured tree dimensions (x1, x2, x3, …) to above-ground biomass. Tree dimensions are measured in sample plot p of stratum i at a given point of time in year t. Tree dimensions x1, x2, x3, … could be, for example DBH, height of tree, etc.
R_j	Root-shoot ratio for tree species j; dimensionless
$B_{TREE,p,i,t}$	Tree biomass in sample plot p in stratum i at a given point of time in year t; t d.m.
$b_{TREE,p,i,t}$	Tree biomass per hectare in sample plot p in stratum i at a given point of time in year t; t d.m. ha
$A_{p,i}$	Area of sample plot p in stratum i; ha
$b_{TREE,i,t}$	Mean tree biomass per hectare in stratum i at a given point of time in year t; t d.m. ha-1
n_i	Number of sample plots in stratum i
s_i^2	Variance of tree biomass per hectare in stratum i at a given point of time in year t; (t d.m. ha-1)^2
$b_{TREE,t}$	Mean tree biomass per hectare within the project boundary at a given point of time in year t; t d.m. ha^{-1}

w_i	Ratio of the area of stratum i to the sum of areas of biomass estimation strata; dimensionless
$s^2_{bTREE,t}$	Variance of mean tree biomass per hectare within the project boundary at a given point of time in year t; (t d.m. ha-1)^2
M	Number of tree biomass estimation strata within the project boundary
$u_{bTREE,t}$	Uncertainty of tree biomass per hectare within the project boundary at a given point of time in year t; %
t_{VAL}	Two-sided Student's t-value for: (i) Degrees of freedom equal to n – M, where n is total number of sample plots within the project boundary, and M is the total number of tree biomass estimation strata; and (ii) a confidence level of 90%.
$s_{bTREE,t}$	Square root of the variance of mean tree biomass per hectare within project boundary at a given point of time in year t (i.e. the standard error of the mean); t d.m. ha^{-1}
$C_{TREE,t}$	Carbon stock in tree biomass within the project boundary at a given point of time in year t; t CO_2-e
$B_{TREE,t}$	Total tree biomass within the project boundary at a given point of time in year t; t d.m.
CF_{TREE}	Carbon fraction of tree biomass; t C t d.m.$^{-1}$. A default value of 0.47 is used unless transparent and verifiable information can be provided to justify a different value.
$dC_{TREE,(t1,t2)}$	Rate of change in carbon stock in tree biomass within the project boundary during the period between a point of time in year t1 and a point of time in year t 2; t CO_2-e yr^{-1}
$C_{TREE,t2}$	Carbon stock in tree biomass within the project boundary at a point of time in year t_2; t CO2-e
$C_{TREE,t1}$	Carbon stock in tree biomass within the project boundary at a point of time in year t_1; t CO2-e
T	Time elapsed between two successive estimations ($T=t_2 – t_1$); yr
$\Delta C_{TREE,t}$	Change in carbon stock in tree biomass within the project boundary in year t; t CO_2
j	1, 2, 3, … tree species in plot p
p	1, 2, 3, … sample plots in stratum i
i	1, 2, 3, … tree biomass estimation strata within the project boundary
t	1, 2, 3, … years counted from the start of the A/R project activity

Change in carbon stock in shrub biomass in project

Carbon stock in shrub biomass within the project boundary at a given point of time in year t is calculated in accordance with the tool "Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities[78] using the equations 9-12 described before in the section 4.1.

According the applied tool for the first verification, the variable $C_{SHRUB,t1}$ in equation 11 is assigned the value of carbon stock in the shrub biomass at the start of the A/R project activity.

Change in carbon stocks in soil organic carbon in project

Soil organic carbon is set to zero for first project instance as the applicability conditions of the tool "Tool for estimation of change in soil organic carbon stocks due to the implementation of A/R CDM project activities" [79] is not fulfilled for the first project instance (see the section 2.2 of the PD).

Change in carbon stocks in dead wood and litter in project

The carbon stock in dead wood and litter are not selected (See table 6 of the PD) and thus, in accordance with the applied methodology, these pools are set to zero.

4.3 Leakage

According the applied methodology the only leakage emissions that can occur are the GHG emissions due to displacement of pre-project agricultural activities. Procedure to be used for calculation of leakage is described in AR-ACM0003 methodology (Version 01.0.0) under the section 5.6 "Leakage". Leakage emissions are estimated as follows:

$$LK_t = LK_{AGRIC,t} \quad \text{(equation 26)}$$

Where:

LK_t	GHG emissions due to leakage, in year t; tCO_2-e
LK_{AGRICt}	Leakage due to the displacement of agricultural activities in year t, as estimated in the tool "Estimation of the increase in GHG emissions attributable to displacement of pre-project agricultural activities in A/R CDM project activity[80]"; tCO_2-e

As described in paragraph "c) Verification of project emissions and leakage emissions" of section 3.3. of this document the leakage emissions can be considered insignificant for the first monitoring period. Therefore LK_t = 0.

[78] AR-TOOL14. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-14-v3.0.0.pdf
[79] EB 60, Annex 12. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-16-v1.1.0.pdf
[80] EB 51, Annex 15. http://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-15-v1.pdf

4.4 Summary of GHG Emission Reductions and Removals

According the applied methodology the net anthropogenic GHG removals by sinks are calculated as follows:

$$\Delta C_{AR-CDM,t} = \Delta C_{ACTUAL,t} - \Delta C_{BSL,t} - LK_t \quad \text{(equation 27)}$$

Where:

$\Delta C_{AR\text{-}CDM,t}$ Net anthropogenic GHG removals by sinks, in year t; tCO_2-e

$\Delta C_{ACTUAL,t}$ Actual net GHG removals by sinks, in year t; tCO_2-e

$\Delta C_{BSL,t}$ Baseline net GHG removals by sinks, in year t; tCO_2-e

LK_t GHG emissions due to leakage, in year t; tCO_2-e

The results for the first verification are shown in table 5 below.

Table 5. Net GHG Emission removals and VCUs

Year	Baseline net GHG removals (tCO_2e)	Actual net GHG removals (tCO_2e)	GHG emissions due to Leakage (tCO_2e)	Net GHG emission removals (tCO_2e)	Emission Reductions to Buffer Pool (37%) (tCO_2e)	VCUs
03/06/2009-02/06/2010	0	-26.2	0	-26.2	(-9.70) 0	(-16) 0
03/06/2010-02/06/2011	0	262.8	0	262.8	97.2	166
03/06/2011-02/06/2012	0	262.8	0	262.8	97.2	166
03/06/2012-02/06/2013	0	262.8	0	262.8	97.2	166
Total	0	762	0	762	282	480

5 ADDITIONAL INFORMATION

The complete *ex post* calculations are presented in the Excel-file called "Calculations ex post first instance".